

# Mixing-Space Magnetic Particle Imaging

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**Abstract**—In this article, we propose a novel approach of mixing-space magnetic particle imaging (MPI) based on the combination of the spatial domain and frequency domain. For 2-D MPI, the superparamagnetic iron oxide nanoparticles (SPIONs) spectra are measured for frequency-domain reconstruction in the  $x$ -direction, whereas the SPION spectra in different field-free-point (FFP) positions are used for spatial-domain reconstruction in the  $y$ -direction. The system matrix is formulated with the SPION spectra in different  $x$  positions along the  $x$ -direction and different FFP positions along the  $y$ -direction, thus allowing to deblur the SPION image in both  $x$ - and  $y$ -directions. A 2-D MPI system is designed and built to generate a drive field in the  $x$ -direction and a shift field in the  $y$ -direction, as well as to measure the SPION spectra. Phantom experiments of two spots demonstrate that the proposed approach allows for spatial resolutions of 0.3 mm in the  $x$ -direction and 0.8 mm in the  $y$ -direction at 1.14 T/m. The SPION samples with different iron concentrations are used to assess the limit of detection (LOD), showing an LOD of 1.04  $\mu\text{g}$  (Fe)/mL (104 ng Fe weight) using Perimag SPIONs. The proposed mixing-space MPI approach significantly reduces both the hardware complexity and the processing time for system matrix measurement but still yields good spatial resolution.

**Index Terms**—Limit of detection (LOD), magnetic particle imaging (MPI), mixing space, spatial resolution, superparamagnetic iron oxide nanoparticles (SPIONs).

## I. INTRODUCTION

MAGNETIC particle imaging (MPI) is a new imaging modality that has shown great promise in biomedical applications [1], [2], including vascular imaging [3], [4], cell tracking [5], [6], [7], drug delivery [8], and magnetic hyperthermia [9]. As a tracer-based method, MPI provides a unique combination of features, such as quantification, high spatial and temporal resolution, high sensitivity, and free of ionizing radiation [10]. In MPI, a dynamic magnetic field is used

to magnetize the superparamagnetic iron oxide nanoparticles (SPIONs), and then, the nonlinear magnetization of SPIONs leads to an induced electromotive force in the receive coil. The measured voltages in the receive coils are utilized to obtain the spatial distribution of magnetic particles [11].

Several imaging parameters are used to quantitatively evaluate the performance of an MPI approach or system, including the field of view (FOV), temporal resolution, spatial resolution, and limit of detection (LOD) [12]. Typically, these parameters are linked to each other. For example, increasing the LOD by averaging the signal will decrease the temporal resolution. Theoretically, these parameters can be improved to a very high level using more coils, a strong gradient field, and high excitation amplitude [13], [14]. However, these means will lead to higher hardware complexity and cost. Therefore, for a given application, an MPI system should be carefully designed by optimizing its hardware complexity while satisfying the performance constraints.

In the last few years, MPI has experienced rapid development. Several MPI systems have been designed and developed based on different approaches. Saritas et al. [15] developed a field free point (FFP) and a field free line (FFL)-based MPI scanners, in which spatial resolution can reach  $\sim 2$  mm at 7 T/m and  $\sim 6$  mm at 2.4 T/m, respectively. Magnetic Insight Inc. developed a commercial scanner that has an FOV of  $60 \times 60 \times 120$  mm [16]. Tay et al. [17] reported the performance of super-ferromagnetic nanoparticles in the MPI and achieved a spatial resolution of 0.15 mm. Graeser et al. [18] developed a human-sized MPI scanner that allows for an LOD of 263 pmol Fe/mL. The scanner comprises one drive coil, two selection coils, and two receive coils. The image is reconstructed using a joint approach, which only uses the frequency-domain information [19]. Vogel et al. [20], [21] proposed an approach called “traveling-wave MPI” to expand FOV and achieved a high imaging speed up to 2000 frames/s and a spatial resolution of about 2–3 mm. The traveling-wave MPI scanner comprises an array of coils to generate a dynamic linear gradient array, which requires high hardware complexity. Le et al. [22] presented a design guideline for a 3-D large FOV MPI scanner, which has an FOV of  $186 \times 172 \times 122$  mm with a gradient of 2 T/m/ $\mu_0$ . The overall 3-D amplitude-modulation-based MPI scanner comprises one receive coil, five drive coils, and two selection coils. A summary of the major reported MPI scanners is shown in Table I.

The imaging process of the above researches can be roughly divided into frequency-domain-based and spatial-domain-based approaches. Each method has its pros and cons. The frequency-domain approach uses a system matrix that contains rich harmonic information of the magnetization

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TABLE I  
COMPARISON OF MAJOR REPORTED MPI SYSTEMS

| Research Institutions                       | Device                              | Field of View                                                   | Temporal Resolution | Spatial Resolution                                            | Limit of Detection                    |
|---------------------------------------------|-------------------------------------|-----------------------------------------------------------------|---------------------|---------------------------------------------------------------|---------------------------------------|
| Magnetic Insight                            | Magnetic Insight Momentum           | $\sim 60 \text{ mm} \times 60 \text{ mm} \times 120 \text{ mm}$ | —                   | 0.6 mm @ 6 T/m                                                | 550 ng Fe/mL                          |
| University of California Berkeley           | Self-Developed FFP MPI              | $\sim 40 \text{ mm} \times 60 \text{ mm}$                       | —                   | 0.15 mm @ 6.3 T/m                                             | —                                     |
| University Hamburg                          | Self-Developed Human Brain Size MPI | $100 \text{ mm} \times 140 \text{ mm}$                          | 26 s                | X: 5 mm @ 0.1 T/m<br>Y: 6 mm @ 0.2 T/m<br>Z: 20 mm @ 0.1 T/m  | 2 $\mu\text{g}$ Fe @ 50 $\mu\text{L}$ |
| University of Würzburg                      | Traveling Wave MPI                  | $65 \text{ mm} \times 29 \text{ mm}$                            | 500 $\mu\text{s}$   | 2 mm - 3 mm                                                   | —                                     |
| Gwangju Institute of Science and Technology | Self Developed Human Size MPI       | $90 \text{ mm} \times 90 \text{ mm} \times 90 \text{ mm}$       | 1 s @ 2-D           | X: 7 mm @ 2.5 T/m<br>Y: 5 mm @ 1.25 T/m<br>Z: 8 mm @ 1.25 T/m | 5 $\mu\text{g}$ Fe @ 50 $\mu\text{L}$ |
| Beihang University                          | 2-D Mixing-Space MPI                | $30 \text{ mm} \times 30 \text{ mm}$                            | 1 s                 | X: 0.3 mm @ 1.2 T/m<br>Y: 0.8 mm @ 1.2 T/m                    | 104 ng Fe @ 100 $\mu\text{L}$         |

response. It has a higher temporal resolution and is more time-consuming for the measurements of the system matrix compared with the spatial-domain approach. However, it relies on the complex configuration of the drive coils and excitation coils to obtain complex trajectories, and the inverse problem is usually ill-posed. The spatial-domain approach uses the measured point spread function (PSF) or FFP positions. Taking the narrowband MPI as an example, the spatial-domain approach needs to obtain the magnetization signal when the FFP is in each pixel. Another spatial-domain approach, named  $x$ -space MPI, is a specific type of reconstruction where the acquired signal is directly gridded, and dc recovery is then applied. This method requires a one-time amplitude and phase calibration [23], [24]. The image is reconstructed by gridding the received signal over time to pixels. The inverse problem is well-posed but time-consuming to record the movement of the FFP through all grids on the FOV.

In this article, we propose a novel mixing-space MPI approach based on the spatial-domain information and frequency-domain information. The main contribution of this article is described as follows.

- 1) A mixing-space encoding method is proposed by combining the spatial-domain and frequency-domain information, which improves the image quality.
- 2) A new measurement method of the system matrix for mixing-space MPI is proposed to decrease the acquisition time of the system matrix.
- 3) A mixing-space MPI system is designed and built. The device comprises two permanent magnets, one shift coil, one drive coil, and one receive coil. Experimental assessment of the device demonstrates that it yields very good performance while requiring light hardware.

## II. METHODOLOGY

### A. Principle of Mixing-Space MPI

In this study, a mixing-space MPI approach is investigated based on the combination of the frequency domain and spatial

domain. A drive magnetic field (DMF) is applied to excite the SPIONs and scan the FFP through the FOV in the  $x$ -direction, whereas a quasi-static magnetic field is applied to monitor the FFP through the  $y$ -direction. Thus, the spectra of SPIONs at different  $x$  positions are measured for encoding and imaging in the  $x$ -direction, whereas the spectra dependent on the position of FFP in the  $y$ -direction are measured for imaging in the  $y$ -direction. The principle of the 2-D mixing-space MPI approach is depicted in Fig. 1. A pair of permanent magnets are arranged in the Maxwell configuration for the generation of the gradient magnetic field  $\mathbf{H}_G$ , as well as the FFP. To achieve spatial encoding, the FFP can be moved by the shift field  $\mathbf{H}_y$  in the  $y$ -direction and the drive field  $\mathbf{H}_x$  in the  $x$ -direction, as shown in Fig. 1(a).

The strength of  $\mathbf{H}_y$  in the  $y$ -direction is shown in Fig. 1(b). The position of the FFP in the  $y$ -direction is equal to  $-H_y/G_y$ , where  $G_y$  denotes the gradient of the static magnetic field in the  $y$ -direction. The shift field is segmented into short-time subinterval  $dt$ , as shown in Fig. 1(b). The movement of FFP in the  $y$ -direction is shown in Fig. 1(c). The frequency  $f_y$  of the shift field  $H_y$  is much lower than the frequency  $f_x$  of the drive field  $H_x$ . Therefore, the position of FFP in the  $y$ -direction within one subinterval is considered unchanged. In one subinterval, the drive field is shown in Fig. 1(d). The principle of SPION magnetization under shift and drive fields is shown in Fig. 1(e). The schematic of the frequency-domain signal is shown in Fig. 1(f).

Without considering the particle relaxation, the SPION magnetization  $M(t)$  can be described by the static Langevin function [25]

$$M(t) = n\bar{m}\mathcal{L}(\xi) = n\bar{m}[\coth(\xi) - 1/\xi] \quad (1)$$

where the Langevin parameter  $\xi = \mu_0 m \mathbf{H} / k_B T$  is the ratio of magnetic energy to thermal energy,  $n$  denotes the number of SPIONs,  $\bar{m}$  denotes the mean magnetic moment of SPIONs,  $\mu_0$  denotes the permeability of free space,  $k_B$  denotes Boltzmann's constant,  $T$  denotes the particle temperature, and  $\mathbf{H}$  denotes the magnetic field. In the FOV with gradient dc, shift,

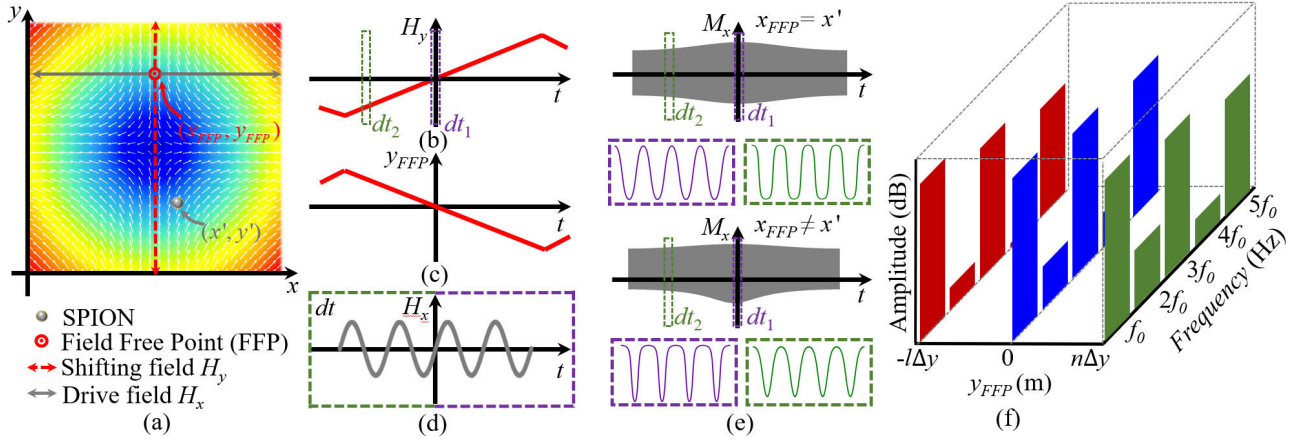


Fig. 1. Basic principle of the 2-D mixing-space MPI. (a) Schematic of magnetic field configuration. (b) Schematic of shift field. (c) Movement of FFP in  $y$ -direction. (d) Schematic of drive field. (e) Principle of SPION magnetization under shift and drive field. (f) Schematic of the frequency-domain signal.

and DMFs, the magnetization  $\vec{M}_0(x, y, t)$  of a spot SPION sample at the position  $(x, y)$  is expressed approximately by [26]

$$\begin{aligned} \vec{M}_0(x, y, t) &= n\bar{m}\mathcal{L}\left(\frac{\mu_0\bar{m}\mathbf{H}(x, y, t)}{k_B T}\right) \\ &= n\bar{m}\mathcal{L}(\beta(\mathbf{H}_G(x, y) + \mathbf{H}_x(t) + \mathbf{H}_y(t))) \\ &= n\bar{m}\mathcal{L}(\beta[G_x \cdot x \cdot \vec{x} + G_y \cdot y \cdot \vec{y} + H_x(t) \cdot \vec{x} + H_y(t) \cdot \vec{y}]) \end{aligned} \quad (2)$$

where  $\beta = (\mu_0\bar{m})/(k_B T)$ .

For different subintervals  $dt$ , the difference (DIFF) in shift field strength leads to the different starting positions of the magnetization curves. The total magnetization response of SPIONS placed in different  $x$ -axis positions is shown in Fig. 1(e). Due to the nonlinear magnetization characteristics, the magnetization of SPIONS in different subintervals contains different harmonic information. The variation of the spectra with the position of  $y_{FFP}$  is shown in Fig. 1(f).

With the SPION concentration distribution  $c(x, y)$ , the magnetization of the phantom in one subinterval can be expressed as follows:

$$\vec{M}(t) = \sum_x \sum_y c(x, y) \cdot \vec{M}_0(x, y, t). \quad (3)$$

The voltage induced in the receive coil is given by Faraday's law [27]

$$\begin{aligned} U(t) &= -\frac{d}{dt} \int_V \vec{M}(t) \cdot p dV \\ &= -\frac{d}{dt} \int_V \left[ \sum_x \sum_y c(x, y) \cdot \vec{M}_0(x, y, t) \right] \cdot p dV \\ &= \sum_x \sum_y c(x, y) \cdot -\frac{d}{dt} \int_V \vec{M}_0(x, y, t) \cdot p dV \\ &= \sum_x \sum_y c(x, y) \cdot u(x, y, t) \end{aligned} \quad (4)$$

where  $U$  denotes the total induced voltage,  $p$  denotes the sensitivity of the receive coil, and  $u$  denotes the voltage induced by the magnetization  $\vec{M}_0$ .

After the discretization of space, (4) can be described by a linear equation

$$U = \sum_{m=-M}^M \sum_{n=-N}^N c(x_m, y_n) \cdot u(x_m, y_n) \quad (5)$$

where  $2M+1$  and  $2N+1$  denote the number of pixels in the  $x$ -direction and  $y$ -direction, respectively;  $U$  denotes the measured induced voltage of the phantom; and  $u(x, y)$  denotes the induced voltage when an SPION sample is placed at position  $(x, y)$ .

By analyzing the spectrum, the voltages  $U$  and  $u$  are transformed to  $\hat{U}$  and  $\hat{u}$ . Then, after the discretization of time,  $\hat{U}$  can be expressed as follows:

$$\hat{U} = \begin{bmatrix} \hat{U}^{-l} & \hat{U}^{-l+1} & \dots & \hat{U}^l \end{bmatrix}_{((2l+1) \cdot K, 1)}^T \quad (6)$$

where  $\hat{U}^l$  denotes  $\hat{U}$  at the  $l$ th subinterval in which  $y_{FFP}$  is equal to  $y_l$  and  $l \in \{-N, N\}$ , in which  $l$  denotes the number of measurement subinterval. Thus,  $\hat{U}^l$  can be given by

$$\hat{U}^l = \begin{bmatrix} U_1^l & U_2^l & \dots & U_K^l \end{bmatrix}_{(K, 1)}^T \quad (7)$$

where  $U_k^l$  denotes the  $k$ th harmonic of  $\hat{U}^l$  and  $k \in \{1, K\}$ , in which  $K$  is the number of selected harmonics.

$\hat{u}(x, y)$  can be detailed as

$$\begin{aligned} \hat{u}(x, y) &= \begin{bmatrix} \hat{u}^{-L}(x, y) & \hat{u}^{-L+1}(x, y) & \dots & \hat{u}^L(x, y) \end{bmatrix}_{((2L+1) \cdot K, 1)}^T \end{aligned} \quad (8)$$

where  $\hat{u}^l(x, y)$  denotes  $\hat{u}(x, y)$  at the  $l$ th subinterval in which  $y_{FFP}$  is equal to  $y_l$ .  $\hat{u}^l(x, y)$  can be given by

$$\hat{u}^l(x, y) = \begin{bmatrix} u_1^l(x, y) & u_2^l(x, y) & \dots & u_K^l(x, y) \end{bmatrix}_{(K, 1)}^T \quad (9)$$

where  $u_k^l(x, y)$  denotes the  $k$ th harmonic of  $\hat{u}^l(x, y)$ .

### B. Image Reconstruction

At a given FFP position, the forward model for 1-D imaging of a line ( $y - y_{\text{FFP}}$ ) in the  $x$ -direction can be simply expressed as

$$\mathbf{A}(y - y_{\text{FFP}}) \cdot \mathbf{c}(y - y_{\text{FFP}}) = \hat{\mathbf{U}}(y - y_{\text{FFP}}) \quad (10)$$

where  $y_{\text{FFP}}$  denotes the position of FFP in the  $y$ -direction,  $\mathbf{A}(y - y_{\text{FFP}})$  denotes the system matrix,  $\mathbf{c}(y - y_{\text{FFP}})$  denotes the SPION distribution in the  $x$ -direction, and  $\hat{\mathbf{U}}(y - y_{\text{FFP}})$  denotes the measured spectra of the phantom. The system matrix can be expressed as

$$\begin{aligned} & \mathbf{A}(y - y_{\text{FFP}}) \\ &= \begin{bmatrix} u_1^0(x_{-M}, y_{\text{DIFF}}) & \dots & u_1^0(x_M, y_{\text{DIFF}}) \\ u_2^0(x_{-M}, y_{\text{DIFF}}) & \dots & u_2^0(x_M, y_{\text{DIFF}}) \\ \vdots & \ddots & \vdots \\ u_K^0(x_{-M}, y_{\text{DIFF}}) & \dots & u_K^0(x_M, y_{\text{DIFF}}) \end{bmatrix}_{(K, 2M+1)} \end{aligned} \quad (11)$$

where  $\text{DIFF} = (y - y_{\text{FFP}})/\Delta y$  and  $\text{DIFF} \in \{-2N, 2N\}$ ,  $\Delta y$  denotes the space of the pixels in the  $y$ -direction,  $u_k(x, y)$  denotes the  $k$ th harmonic of induced voltage when a spot SPION sample is placed at position  $(x, y)$ , and  $2M+1$  denotes the number of pixels in the  $x$ -direction. The SPION distribution can be expressed as

$$\mathbf{c}(y - y_{\text{FFP}}) = [c(x_{-M}, y_{\text{DIFF}}) \quad \dots \quad c(x_M, y_{\text{DIFF}})]^T. \quad (12)$$

The measured spectra of the phantom can be expressed as

$$\hat{\mathbf{U}}(y - y_{\text{FFP}}) = [U_1^{\text{DIFF}} \quad U_2^{\text{DIFF}} \quad \dots \quad U_K^{\text{DIFF}}]^T \quad (13)$$

where  $U_k$  denotes the  $k$ th harmonic of  $\hat{\mathbf{U}}(y - y_{\text{FFP}})$ .

For 2-D imaging, this linear system can be written as [28]

$$\mathbf{A}\mathbf{c} = \hat{\mathbf{U}} \quad (14)$$

where  $\mathbf{A}$  denotes the system matrix, which can be expressed as

$$\mathbf{A} = \begin{bmatrix} \hat{\mathbf{u}}^{-N}(x_{-M}, y_{-N}) & \dots & \hat{\mathbf{u}}^{-N}(x_M, y_N) \\ \hat{\mathbf{u}}^{-N+1}(x_{-M}, y_{-N}) & \dots & \hat{\mathbf{u}}^{-N+1}(x_M, y_N) \\ \vdots & \ddots & \vdots \\ \hat{\mathbf{u}}^N(x_{-M}, y_{-N}) & \dots & \hat{\mathbf{u}}^N(x_M, y_N) \end{bmatrix}. \quad (15)$$

For the mixing-space MPI, the relationship of the element of the system matrix can be expressed as

$$\hat{\mathbf{u}}^l[x_m, y_n] = \hat{\mathbf{u}}^0[x_m, (n-l) \cdot \Delta y] = \hat{\mathbf{u}}^0[x_m, y_{n-l}] \quad (16)$$

where  $m \in [-M, M]$ ,  $n \in [-N, N]$ , and  $l \in [-N, N]$ . Based on the above substitution relationship, the system matrix can be rewritten as

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}(0 \cdot \Delta y) & \dots & \mathbf{A}(2N \cdot \Delta y) \\ \mathbf{A}(-1 \cdot \Delta y) & \dots & \mathbf{A}((2N-1) \cdot \Delta y) \\ \vdots & \ddots & \vdots \\ \mathbf{A}(-2N \cdot \Delta y) & \dots & \mathbf{A}(0 \cdot \Delta y) \end{bmatrix} \quad (17)$$

where  $\mathbf{A}((n-l) \cdot \Delta y)$  denotes the system matrix of 1-D MPI.  $\mathbf{c}$  denotes the SPION concentration distribution of the phantom, which can be expressed as

$$\mathbf{c} = [\mathbf{c}(-N \cdot \Delta y)^T \quad \dots \quad \mathbf{c}(N \cdot \Delta y)^T]^T. \quad (18)$$

Then, the forward model for the 2-D mixing-space MPI can be expressed as

$$\mathbf{A}_{((2N+1) \cdot K, (2M+1) \cdot (2N+1))} \cdot \mathbf{c}_{((2M+1) \cdot (2N+1), 1)} = \hat{\mathbf{U}}_{((2N+1) \cdot K, 1)}. \quad (19)$$

The elements in  $\hat{\mathbf{U}}^l$  are the harmonics of magnetization response at fixed FFP  $y$ -axis positions, which represent the frequency-domain information. The rows of  $\hat{\mathbf{U}}$  are the discretization of the FFP position, representing the spatial-domain information. This forward model contains both the frequency-domain information and spatial-domain information, thus called the mixing-space method.

Based on the above forward model of the mixing-space MPI approach, solving (19) allows one to obtain the particle distribution from the measured voltages. Increasing  $N$  and  $K$  is beneficial for solving (19) and improving the resolution of the output image. The maximum value of  $N$  is  $(f_x/f_y - 1)/2$ , which means that  $dt$  should be at least one period of the drive field. Increasing  $dt$  will improve the SNR of the measured signal by averaging but will decrease  $N$ . The maximum value of  $K$  depends on the SNR of the measured signal. The high-order harmonics have more detailed spatial features and narrower full-width at half-maximum (FWHM). Therefore, high-order harmonics are helpful to improve the spatial resolution of images. However, the low SNR of high-order harmonics can lead to a higher artifact level.

When the above parameters are determined, (19) can be solved using some regularization reconstruction methods, such as Tikhonov method, singular value decomposition (SVD) method [29], algebraic reconstruction technique (ART) method [30], sparse representation method [31], and deep learning [32], to obtain the SPION concentration distribution. The Tikhonov method is selected in this work, and the iteration procedure can be expressed as

$$\mathbf{c}_{s+1} = \mathbf{c}_s - (\mathbf{A}^T \mathbf{A} + \alpha \mathbf{I})^{-1} \mathbf{A}^T (\mathbf{A} \mathbf{c}_s - \mathbf{U}) \quad (20)$$

where  $\alpha$  is the regularization parameter.

### III. MIXING-SPACE MPI SYSTEM

#### A. System Design

Based on the proposed approach, a 2-D mixing-space MPI system is designed and built. It consists of a power drive module, a magnet module, a signal processing module, and an imaging module. The connection diagram for the mixing-space MPI system is shown in Fig. 2.

The drive module contains power amplifiers (PAs), current sensors, and bandpass filter (BPF). The magnet module contains permanent magnets and coils. The signal processing module contains dc power supplies, band-stop filter (BSF), and low noise amplifier (LNA). The imaging module contains the data acquisition (DAQ) cards and a personal computer (PC) to run professional software.

1) *Power Drive Module*: The small signal to be amplified to the PA is provided by the DAQ. The output current of the PA is supplied to the shift coil and drive coil through the LCC resonance circuit, current sensors, and a fifth-order BPF. The LCC resonance circuit is used for power matching, and the



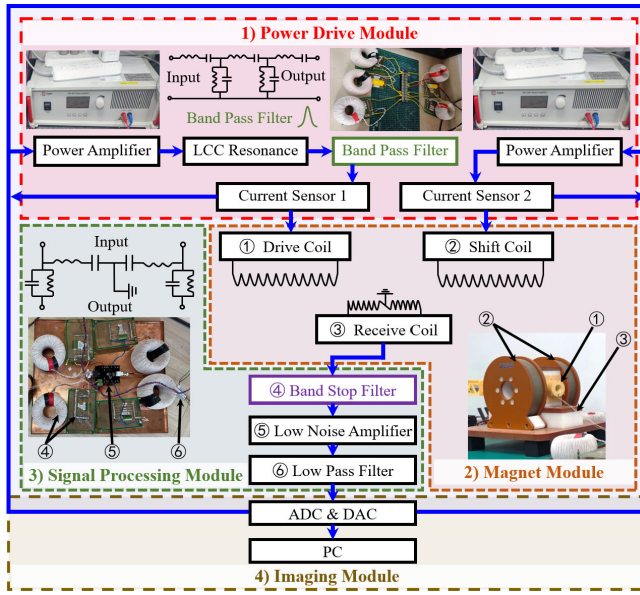


Fig. 2. Connection diagram for the mixing-space MPI system.

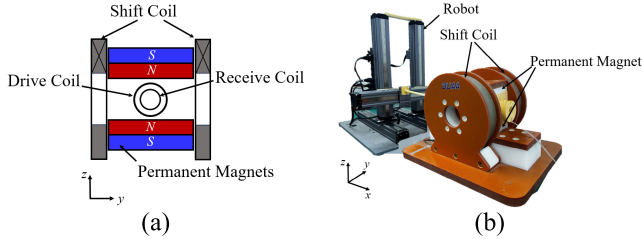


Fig. 3. (a) Schematic of the magnet configuration. (b) Photograph of the MPI system.

BPF is used to reduce the total harmonic distortion. The output of the current sensor is collected by the DAQ to calculate the magnetic field strength.

2) *Magnet Module*: Fig. 3 shows the schematic of the magnet configuration and the photograph of the MPI system. A pair of NdFeB permanent magnets are oppositely arranged in the  $z$ -direction to generate the gradient magnetic field with gradients of 1.1, 1.1, and 2.2 T/m in the  $x$ -,  $y$ -, and  $z$ -directions. A pair of Helmholtz coils are designed in the  $y$ -direction to generate the shift magnetic field for FFP scanning in the  $y$ -direction. A solenoid coil is used to generate the DMF with a frequency of 10 kHz in the  $x$ -direction for FFP scanning in the  $x$ -direction and SPION excitation. A pair of Helmholtz coils are used to generate a 1-Hz triangular-wave magnetic field to scan the FFP in the  $y$ -direction. The receive coil, which is wound in a 3-D-printed resin bobbin, consists of a solenoid detection coil and a compensation coil. The robot is used to move the particle sample in the FOV.

The gradient field strength is measured using a Gauss meter, as shown in Fig. 4. The gradient of the static gradient field in the  $x$ - and  $y$ -directions is 1.144 T/m. The gradient in the  $z$ -direction is 2.238 T/m.

For 2-D mixing-space MPI, the temporal resolution is related to the time consumption to move the FFP in the  $y$ -direction, which is 1 s for the developed system. The temporal resolution can be further improved by increasing the shifting frequency, while the sampling frequency should also

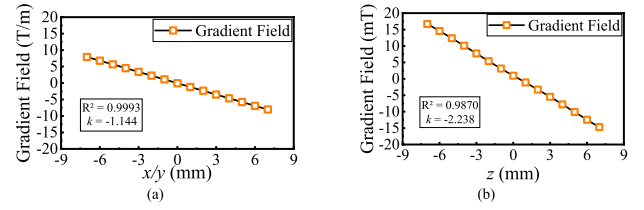


Fig. 4. Measured gradient field using a Gauss meter in (a)  $x$ -direction and  $y$ -direction, and (b)  $z$ -direction.

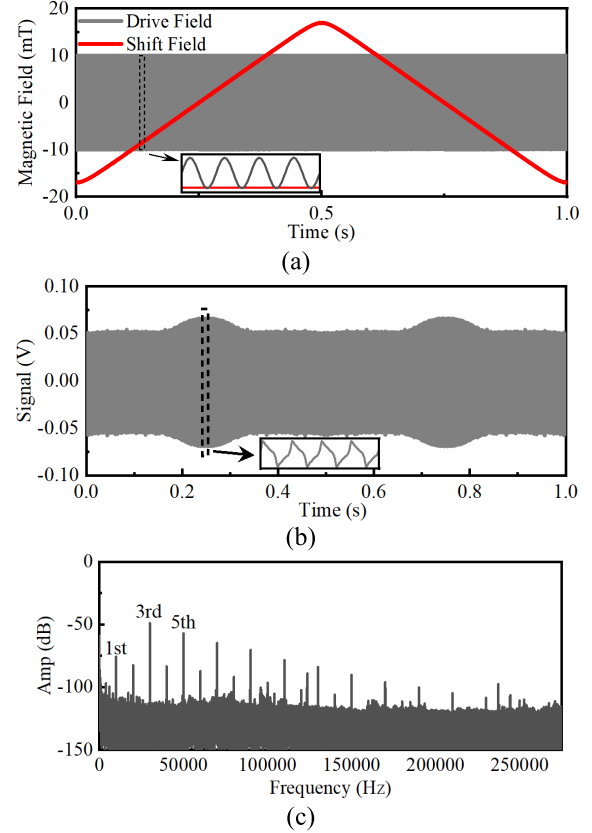


Fig. 5. (a) Measured shift field and drive field. (b) Measured magnetization response and (c) spectrum.

be increased. The magnetic field strength of the shift field and drive field can reach 20 mT, which allows an FOV of  $17 \times 17$  mm.

3) *Signal Processing Module*: The dc power supplies provide electrical energy for current sensors and amplifiers. The received signal is filtered using a second-order BSF to suppress the feed-through and is amplified by an LNA. Then, the 2nd–26th harmonics of the SPION magnetization signal, whose frequencies are in the range of 20–260 kHz, are collected by a DAQ card with a sampling rate of 3 MHz.

4) *Imaging Module*: The amplitude and phase of the harmonic are demodulated using digital lock-in detection technology. The digital signals are processed, and the MPI image is reconstructed on the PC.

The measured shift field and drive field are shown in Fig. 5(a). The measured SPION magnetization response and spectrum are shown in Fig. 5(b) and (c). The SNR of the received signal was tested based on a DAQ card, and the noise floor level is about  $-120$  dB.

### B. Measurement of System Matrix

According to (1), the total induced voltage is an integral of the convolution of the SPION concentration distribution with a Langevin function kernel [33], [34]. Therefore, the Langevin function kernel  $u(x, y, t)$  is assigned to be the PSF of the mixing-space MPI. The PSF  $[x, y]$  can be defined as the induced voltage of the unit concentration point SPIONs sample, which is placed at the position  $(x, y)$  without a shift field [35]. The relationship between PSF and  $u^l[x, y]$  can be expressed as

$$\text{PSF}[x_m, y_n] = u^0[x_m, y_n]. \quad (21)$$

To reconstruct the MPI image using the built system, the PSF should be measured at first. The spatial position of SPIONs needs to be changed to obtain the element  $u_k^j[x, y]$  of the system matrix, which means that it is time-consuming and requires a precision 2-D displacement table.

In this work, a new measurement method of system matrix for mixing-space MPI is proposed to decrease the processing time. In the mixing-space MPI, the SPION position in the  $y$ -direction can be equivalently varied by a shift field with a large amplitude, according to (1). The movement of SPIONs in the  $x$ -direction is achieved by a robot.

For the mixing-space MPI, the relationship of the element of the system matrix and PSF can be expressed as

$$u^l[x_m, y_n] = u^0[x_m, y_{n-l}] = \text{PSF}[x_m, y_{n-l}] \quad (22)$$

where  $m \in [-M, M]$ ,  $n \in [-N, N]$ , and  $l \in [-N, N]$ . Therefore, the information of the complex system matrix can be covered by  $\text{PSF}[x_m, y_n]_{m \in [-M, M], n \in [-2N, 2N]}$ . The measurement time of the system matrix can be greatly reduced by expanding the amplitude of the driving magnetic field, thus expanding the FOV in the  $y$ -direction. The improved measurement time can be calculated by

$$t_{\text{system}} = M \cdot t_{\text{robot}} \quad (23)$$

where  $t_{\text{system}}$  denotes the measurement time of the system matrix,  $M$  denotes the number of pixels in the  $x$ -direction, and  $t_{\text{robot}}$  denotes the displacement time of the robot.

The measurement steps of the system matrix are summarized as follows.

#### Procedure 1 Acquisition of the System Matrix of Mixing-Space MPI

- Step 1:** Set the strength of  $H_x$  and  $H_y$ .
- Step 2:** Put a point SPOIN sample in  $[x_m, 0]_{m=-M}$ .
- Step 3:** Set the strength of  $H_y$  to  $-G_y \cdot y_n$ , generate drive field  $H_x$  and obtain  $\text{PSF}[x_m, y_n]$ ,  $n \in [-2N, 2N]$ .
- Step 4:** Change the position of the point SPOINs sample by the displacement platform in  $x$ -direction, if  $m < M$ ,  $m = m + 1$  and return to step 3, else step 5.
- Step 5:** Convert  $\text{PSF}[x_m, y_n]_{m \in [-M, M], n \in [-2N, 2N]}$  to system matrix.

The measured PSF of the developed system is shown in Fig. 6. The FOVs in the  $x$ - and  $y$ -directions are 15 and 30 mm, respectively. The higher harmonic of the measured signal has

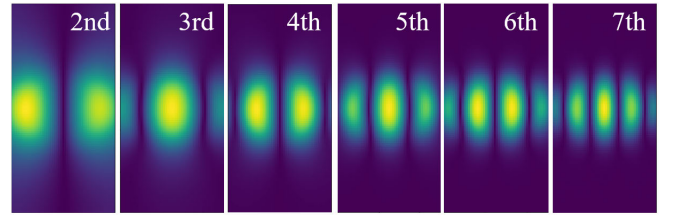


Fig. 6. Measured different harmonics components of PSF under 10-mT drive field.

a lower amplitude and thus lower SNR. However, it has more detailed spatial features that may help improve the spatial resolution of images. In this work, the 2nd–26th harmonics of the measured signal are used in the system matrix.

## IV. MIXING-SPACE MPI SYSTEM

### A. Experimental Description

Experiments were carried out to evaluate the performance of the designed mixing-space MPI system. Plain Perimag<sup>1</sup> SPIONs with a hydrodynamic diameter of 130 nm were used to investigate the spatial resolution and LOD of the system. The iron concentration of the stock solution was 8.5 mg/mL. In the experiments, the strength of the shift field and drive field was set to 17.04 and 10 mT, respectively. The FOV was  $15 \times 15$  mm, and the pixel size was  $0.3 \times 0.3$  mm.

### B. Spatial Resolution

Various phantoms of cylindrical shapes filled with SPIONs according to different layouts were used to evaluate the spatial resolution of the proposed mixing-space MPI. The diameter and depth of the cylinders were 1.5 and 3 mm, respectively. The phantom was filled with 5- $\mu$ L solution of Perimag<sup>1</sup> SPIONs with an iron concentration of 25 mg/mL. The edge-to-edge distance between the two cylinders in the phantom varies from 0.3 to 5 mm. The 2-D images and 1-D comparison reconstructed using the joint 1-D frequency-domain reconstruction and mixing-space reconstruction methods are shown in Fig. 7.

In Fig. 7(a), the distances of the cylinders filled with SPION solution changed from 0.8 to 0.3 mm in the  $x$ -direction. The images reconstructed using the joint frequency-domain method showed good spatial resolution in the  $x$ -direction but were deformed in the  $y$ -direction. The images reconstructed using the proposed mixing-space method had a better match with the real phantoms in both shape and size. The pixel values along the blue and red lines are compared. The modulation transfer function (MTF) is one of the most effective criteria for quantitative evaluation of the spatial resolution in an imaging modality, which is given by

$$\text{MTF} = (c_{\text{max}} - c_{\text{min}}) / (c_{\text{max}} + c_{\text{min}}) \quad (24)$$

where  $c_{\text{max}}/c_{\text{min}}$  is the maximum/minimum value of the curve. In the  $x$ -direction, the spatial resolution of the joint frequency-domain method is about 0.6 mm, and the MTF is 56.16%. Meanwhile, the mixing-space method improves the spatial resolution from 0.6 to 0.3 mm, where MTF is 64.20%. In Fig. 7(b), the distances of the cylinders filled with

<sup>1</sup>Registered trademark.

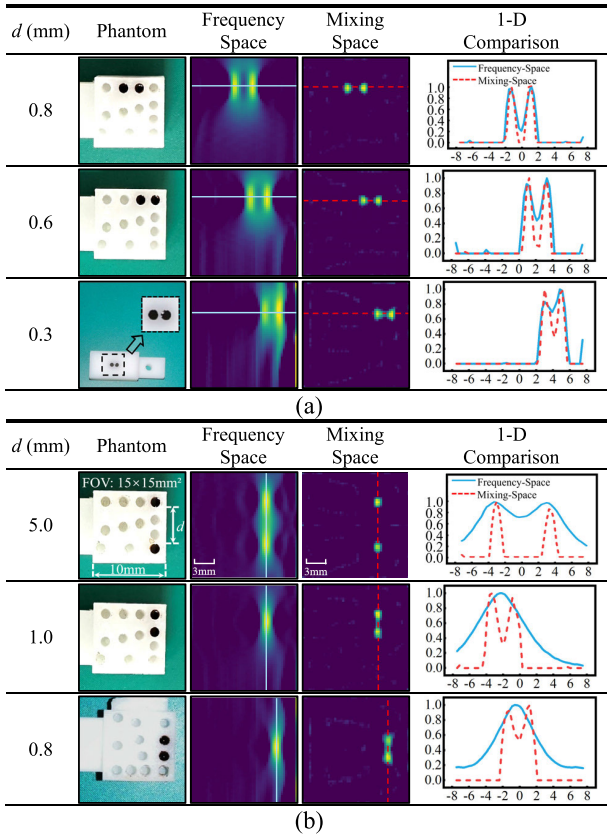


Fig. 7. Photographs of the phantoms, reconstructed images using the joint frequency-domain method, reconstructed images using the mixing-space method, and 1-D comparison of the pixel values between the two methods in (a)  $x$ - and (b)  $y$ -directions.

SPION solution changed from 5 to 0.8 mm in the  $y$ -direction. The images reconstructed using the joint frequency-domain method showed poor image quality in both  $x$ -direction and  $y$ -direction. The images reconstructed using the mixing-space method still had a much better visual effect on shape and size. The pixel values along the blue and red lines are compared. In the  $y$ -direction, the spatial resolution of the joint frequency-domain method is higher than 5 mm. Meanwhile, the mixing-space method improves the spatial resolution to 0.8 mm, where  $\text{MTF} = 47.9\%$ . In the developed MPI system, the drive coil and receive coil are both solenoid coils in the  $x$ -direction. The component of particle magnetization in the  $x$ -direction, which is detected by the receive coil, is more sensitive to the displacement in the  $x$ -direction than that in the  $y$ -direction. Also, the imaging method is different in different directions. Therefore, the spatial resolution in the  $x$ -direction is better than that in the  $y$ -direction. There are several published papers that have found this experimental phenomenon [36], [37].

The magnetization response in one interval also contained the contribution of magnetic nanoparticles (MNPs) placed in the other line. The joint frequency-domain method combines the individual 1-D  $x$ -axis reconstruction results but ignores the coupling between signals from different intervals, which is the main reason that the spatial resolution in the  $y$ -direction using the mixing-space method is much better than that using the joint frequency-domain method. These lead to poor spatial resolution using the joint frequency-domain method,

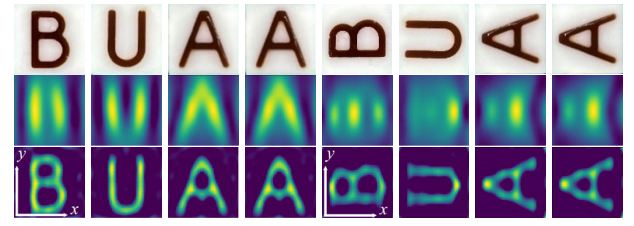


Fig. 8. Photograph of the BUAA-logo phantom filled with SPIONs (line 1), the reconstructed MPI images of joint frequency-domain method (line 2), and mixing-space method (line 3).

TABLE II  
RECONSTRUCTED 2-D IMAGES OF DILUTION SERIES

| Dilution        | 1                 | 10               | 100               | 1000   | 2000   | 4000   | 8000   |
|-----------------|-------------------|------------------|-------------------|--------|--------|--------|--------|
| Iron dose       | 833 $\mu\text{g}$ | 83 $\mu\text{g}$ | 8.3 $\mu\text{g}$ | 830 ng | 415 ng | 208 ng | 104 ng |
| Dilution Series |                   |                  |                   |        |        |        |        |
| Images          |                   |                  |                   |        |        |        |        |

especially in the joint  $y$ -direction [38]. On the other hand, the mixing-space method combines both the frequency-domain information in the  $x$ -direction and spatial-domain information in the  $y$ -direction, which greatly improves the resolution in the  $y$ -direction.

### C. Phantom Reconstruction

To further demonstrate the performance of the proposed mixing-space MPI, a phantom of the logo “BUAA” was reconstructed using the designed MPI system. The size of the phantom is  $10 \times 10$  mm. The phantom was filled with 2.5-mg/mL Perimag<sup>1</sup> SPIONs, as shown in Fig. 8 (line 1). The designed logo was placed in both  $x$ -direction and  $y$ -direction.

### D. Limit of Detection

The limit of detection (LOD) is one of the most important indexes to evaluate the performance of an MPI system. To quantitatively assess the LOD of the custom-built MPI system, a cylinder phantom with a volume of 100  $\mu\text{L}$  was filled with Perimag<sup>1</sup> SPIONs with different iron concentrations ranging from 25 mg/mL to 1.56  $\mu\text{g/mL}$  (iron weight ranging from 850  $\mu\text{g}$  to 104 ng). The dilution series is shown in Table II. The measured FOV was  $15 \times 15$  mm. The strength of the drive field was 10 mT. The reconstructed 2-D images and the fitting curve between the reference concentrations and the max reconstructed value are shown in Table II and Fig. 8, respectively.

In Fig. 9, the red line shows the max reconstructed value of a sample filled with water, which can be considered as the background noise level. The green line shows the theoretical LOD where the  $\text{SNR} > 3$  dB. Fig. 9 shows that the achieved LOD is 104-ng Fe.

### E. Discussion

In this article, the receive coil is set in  $x$ -direction. The spatial resolution can be further improved by using magnetization



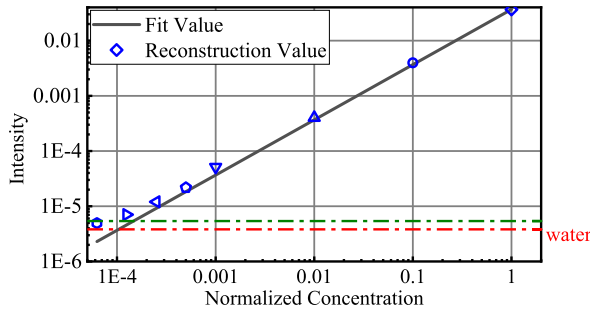


Fig. 9. Fitting curve between the reference concentrations and the max reconstructed value.

response information obtained by multidirectional receive coils. By adding an extra pair of receive coils in the  $y$ -direction, which is more sensitive to the displacement in the  $y$ -direction, the FWHM of the PSF in the  $y$ -direction could be reduced. Therefore, the spatial resolution can be improved with a multidirectional receive coil.

As stated above, the designed MPI scanner allows for an imaging speed of two FPS. It only requires 1-D DMF for the FFP scanning and the SPION excitation in the  $x$ -direction. In the  $y$ -direction, a low-frequency ac magnetic field is generated to scan the FFP. It can be built with thick copper wires with a lot of turns, allowing for higher power efficiency. Thus, the MPI scanner only requires a BSF for drive coil and a BPF for the detection coil in the  $x$ -direction. In addition, due to the significant DIFF in the frequency in the  $x$ - and  $y$ -direction coils, one does not need complex decoupling between  $x$ - and  $y$ -direction coils. It means that all the filter designs are at least a half-costive compared with a frequency-domain based MPI scanner. In addition, the 2-D image is achieved at  $z = 0$ . By adding an extra pair of shifting coils, the FFP can be moved in the  $z$ -direction, and thus, the 2-D MPI can be expanded into 3-D.

## V. CONCLUSION

In this article, a mixing-space MPI approach is proposed by combining the spatial-domain information and frequency-domain information. A mixing-space MPI system is built based on the mixing-space imaging concept. A new measurement method of system matrix for mixing-space MPI system is proposed to reduce the time-consuming. Currently, it takes 5 min to acquire a 2-D system matrix but 1 s to acquire the magnetization response signal for 2-D reconstruction. The experimental results show the following.

- 1) The proposed mixing-space method improves the spatial resolution from 0.6 to 0.3 mm in the  $x$ -direction and from 5 to 0.8 mm in the  $y$ -direction, compared to the traditional joint frequency-domain method.
- 2) The LOD of 104-ng iron is achieved using the built mixing-space MPI system.
- 3) The system matrix can be measured by mechanically shifting a spot sample in the  $x$ -direction and electrically scanning the FFP in the  $y$ -direction, thus significantly decreasing the measurement time.

Considering all the factors involved in MPI, e.g., hardware design efforts, spatial resolution, temporal resolution, sensitivity, and safety, the mixing-space MPI presented in this article

is a very promising approach to reach a good tradeoff in spatial resolution and temporal resolution with lower hardware complexity than in previous works. In addition, the spatial resolution can be further improved by using magnetization response information obtained by multidirectional receive coils.

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